

Titanic Data Cleaning & Preprocessing

Objective

To clean and preprocess the Titanic dataset for machine learning by handling missing values, encoding categorical features, scaling numerical data, and removing outliers.

Tools Used

- Python
- Pandas
- NumPy
- Matplotlib
- Seaborn
- Scikit-learn
- Jupyter Notebook

Dataset

Titanic Dataset

Steps Performed

1. Imported and explored the dataset.
2. Checked data types and missing values.
3. Filled missing values in Age and Embarked columns.
4. Removed the Cabin column due to excessive missing data.
5. Encoded categorical features using Label Encoding.
6. Standardized Age and Fare using StandardScaler.
7. Visualized outliers using boxplots.
8. Removed outliers using the IQR method.
9. Saved the cleaned dataset.

Results

- Missing values were handled successfully.
- Categorical data was converted into numerical format.
- Numerical features were standardized.
- Outliers were identified and removed.
- The dataset was prepared for machine learning models.

Conclusion

Data preprocessing improved the quality of the Titanic dataset by cleaning, transforming, and preparing it for machine learning tasks, leading to more reliable model performance.